NRMD Process Controls System Design Parameters:

**Process Control Requirements:**

The new process control system will have the following functionality:

**System-Wide Components & Alarms**

Low-Level Override System:

- Each tank's low-level pump shutoff interlock shall be equipped with a manual override function.
- A dedicated status indicator light must be provided for each tank to clearly show when the low-level override is active.

Conductivity Monitoring System:

- The system shall continuously monitor water conductivity at critical process points.
- A clear visual display with "Satisfactory" and "Unsatisfactory" status lights must be provided to indicate water quality.
- An "Unsatisfactory" condition should trigger an alarm and will be used as an interlock for PLW to CPW transfers as well as CPW recirc through polishing train.
- The monitoring system must include functionality for alarm testing and silencing.

Automated Process Valve (Solenoid):

- The system uses an automated solenoid valve which will close if conductivity in conductivity meter DMP-CM-1 exceeds a certain level.

**Audible Alarm System & Tank Level Monitoring:**

- Audible alarms shall be installed in both the control room and the main process area to alert personnel to off-normal conditions.
- Alarms will activate for the following events:
    - High-Level Alarm: Tanks reaching a 90% level, as measured by a tank level indicator.

    - Low-Level Alarm: Tanks reaching a 5% level, as measured by a tank level indicator.
    - Tank Full / High-High Level Alarm: Tanks reaching 100% capacity, detected by a separate float switch that serves as a distinct input from the level indicator. This condition will provide a hard interlock to prevent overfilling.
    - Unsatisfactory water conductivity readings.
- A mechanism to silence an active alarm must be available to the operator.

**Power Monitoring System:**

    - There will be a power dampening system installed between the source power and the power that enters the PLC enclosure.

**SMP-P-1 Pump Operations**

General Requirement: The pump will not run if any system tank is at 100% capacity (as detected by a dedicated float switch).

- Transfer from Sump Tank to RLW Tank:
    - The pump's safety disconnect is closed.
    - The pump enable switch is on.
    - The pump mode is set to "Transfer to RLW".
    - The RLW destination tank is selected.
    - The destination tank is not at 90% capacity (as measured by the tank level indicator).
- Polybottle Pumpdown to Sump Tank:
    - The pump's safety disconnect is closed.
    - The pump enable switch is on.
    - The pump mode is set to "Polybottle Pumpdown".
    - The sump tank is not at 90% capacity (as measured by the tank level indicator).
- Recirculation of Sump Tank:
    - The pump's safety disconnect is closed.
    - The pump enable switch is on.
    - The pump mode is set to "Recirc".
- Priming of PPD Pump:
    - The pump's safety disconnect is closed.
    - The pump enable switch is on.
    - The pump mode is set to "PPD Prime

**PPD-P-1 Pump Operations**

General Requirement: The pump will not run if any system tank is at 100% capacity (as detected by a dedicated float switch).

- Transfer from Sump Tank to RLW Tank:
    - The pump's safety disconnect is closed.
    - The pump enable switch is on.
    - The pump mode is set to "Sump".
    - The RLW destination tank is selected.
    - The sump tank is above its 5% low-level setpoint (as measured by the tank level indicator) or the low-level override is engaged.
    - The destination tank is not at 90% capacity (as measured by the tank level indicator).
- Transfer from Pre-Process to RLW/PLW Tank:
    - The pump's safety disconnect is closed.
    - The pump enable switch is on.
    - The pump mode is set to "Pre-Process".
    - A destination tank (RLW or PLW) is selected.
    - The destination tank is not at 90% capacity (as measured by the tank level indicator).

**RLW-P-1 & RLW-P-2 Pump Operations**

General Requirement: The pumps will not run if any system tank is at 100% capacity (as detected by a dedicated float switch).

- Tank-to-Tank Transfer (within RLW or to PLW):
  - The pump's safety disconnect is closed.
  - The pump enable switch is on.
  - The source tank is selected.
  - The destination tank is selected (when transferring to PLW).
  - The source tank is above its 5% low-level setpoint (as measured by the tank level indicator) or the low-level override is engaged.
  - The destination tank is not at 90% capacity (as measured by the tank level indicator).
- Recirculation of RLW Tank:
  - The pump's safety disconnect is closed.
  - The pump enable switch is on.
  - The tank to be recirculated is selected as both the source and destination.
  - The source tank is above its 5% low-level setpoint (as measured by the tank level indicator) or the low-level override is engaged.

**PLW-P-1 & PLW-P-2 Pump Operations**

General Requirement: The pumps will not run if any system tank is at 100% capacity (as detected by a dedicated float switch).

- Tank-to-Tank Transfer (within PLW, to RLW, or to CPW):
    - The pump's safety disconnect is closed.
    - The pump enable switch is on.
    - The source tank is selected.
    - The destination tank is selected (when transferring to RLW or CPW).
    - The source tank is above its 5% low-level setpoint (as measured by the tank level indicator) or the low-level override is engaged.
    - The destination tank is not at 90% capacity (as measured by the tank level indicator).
    - If transferring to CPW tanks, the conductivity monitor must indicate "Satisfactory" and the associated solenoid valve must be open.
- Recirculation of PLW Tank:
    - The pump's safety disconnect is closed.
    - The pump enable switch is on.
    - The tank to be recirculated is selected as both the source and destination.
    - The source tank is above its 5% low-level setpoint (as measured by the tank level indicator) or the low-level override is engaged.

**CPW-P-1 & CPW-P-2 Pump Operations**

General Requirement: The pumps will not run if any system tank is at 100% capacity (as detected by a dedicated float switch) or if the CPW Emergency Stop is activated.

- Tank-to-Tank Transfer (within CPW or to PLW):
  - The pump's safety disconnect is closed.
  - The pump enable switch is on.
  - The source tank is selected.
  - The destination tank is selected (for transfers to PLW).
  - The source tank is above its 5% low-level setpoint (as measured by the tank level indicator) or the low-level override is engaged.
  - The destination tank is not at 90% capacity (as measured by the tank level indicator).
- Transfer to Trailer:
  - The pump's safety disconnect is closed.
  - The pump enable switch is on.
  - The source tank is selected.
  - The mode is set to "Trailer Transfer".
  - The source tank is above its 5% low-level setpoint (as measured by the tank level indicator) or the low-level override is engaged.
- Recirculation of CPW Tank (Polishing):
  - The pump's safety disconnect is closed.
  - The pump enable switch is on.
  - The tank to be recirculated is selected as both the source and destination.
  - The source tank is above its 5% low-level setpoint (as measured by the tank level indicator) or the low-level override is engaged.

**Hardware design parameters:**

The currently existing major system components are the following: 8 pumps, 7 tanks, 9 flow meters, 7 Tank Level indicators, 5 conductivity meters, 1 electrically controlled solenoid valve, 2 process control stations, 1 Cutler Hammer D300 CPU, 2 Eaton D300PSU115 power supplies, 8 control relays, 12x 16 connection point digital input cards, 4x 32 connection point digital output cards.

The following equipment will be replaced during the work of the contract: 9 flow meters, 2 process control stations, 1 Cutler Hammer D300 PLC, 2 Eaton D300PSU115 power supplies, 8 control relays, 12x 16 connection point analog input cards, 4x 32 connection point analog output cards, 2 process control stations, PLC hardware enclosure. Equipment will be replaced with either the units specified below, or with equipment that is functionally identical, if the requested equipment is no longer in active production.

Flowmeter: The currently installed dual channel flowmeters will be removed and replaced with single channel Siemens Flowmeters 1010NR-T2KGS-S2.

CPU: The CPU will be removed and replaced with Allen Bradeley Logix 5562. Currently there is one CPU, in the new design, there will be at least 2 CPU's, with one main and one secondary CPU. In the event of main CPU failure, the system should be designed to immediately switch to the secondary CPU to ensure continuity of operation.

Power Supply: The power supplies will be removed and replaced with 1606XLS240E.

Input Cards: The input cards will be removed and replaced with Allen Bradely. 1756-IB16 for discrete input cards, and 1756-IF16 for analog input cards. Contractor will be responsible for determining required number of input cards.

Output Cards: The output cards will be removed and replaced with Allen Bradely 1756-OB16I output cards. Contractor will be responsible for determining required number of output cards.

The Tank Level Indicators, the pumps, the solenoid valve, the tanks, and the conductivity cells will not be replaced by the work of this contract.

The conductivity meters will be removed and replaced with Rosemount Analytical Inc Conductivity meter model # 1056-01-20-30-AN.

The Process Control Stations will have various switches and buttons. The specific brand/model of switches and buttons will be left to vendor discretion. The process control stations will be organized by instrument (i.e. all buttons and switches related to a pump or tank are in the same general area). The tables below show the required control instrumentation for each Process Control Station :

| Process Control Station 1 | |
|---|---|
| Instrument | Required Control Instrumentation Hardware |
| PPD pump | 1x Start button, 1x Stop button, 5x2 option switch, 1x Running light |
| Sump pump | Start button, Stop button, 1x 3 option switch, 3x 2 option switch,1x speed potentiometer for Variable Frequency Drive, 1x Running light |
| RLW Pump 1 | Start button, Stop button, 4x 2 option switch, 1x Running light |
| RLW Pump 2 | Start button, Stop button, 4x 2 option switch, 1x Running light |
| Sump Tank | 3x push button, 4x 2 option switch |
| RLW Tank 1 | 3x push button, 4x 2 option switch |
| RLW Tank 2 | 3x push button, 4x 2 option switch |
| PLW Tank 1 | 2x push buttons, 2x 2 option switch |
| PLW Tank 2 | 2x push buttons, 2x 2 option switch |

| Process Control Station 2 | |
|---|---|
| Instrument | Required Control Instrumentation |
| PLW pump 1 | 1x Start button, 1x Stop button, 1x 4 option switch, 2x 2 option switch, 2x 3 option switch 1x Running light |
| PLW pump 2 | 1x Start button, 1x Stop button, 1x 4 option switch, 2x 2 option switch, 2x 3 option switch 1x Running light |
| CPW Pump 1 | Start button, Stop button, 4x 2 option switch |
| CPW Pump 2 | Start button, Stop button, 4x 2 option switch |
| RLW Tank 1 | 2x push button, 2x 2 option switch |
| RLW Tank 2 | 2x push button, 2x 2 option switch |
| PLW Tank 1 | 3x push button, 4x 2 option switch |
| PLW Tank 2 | 3x push button, 4x 2 option switch |
| CPW Tank 1 | 4x push buttons, 5x 2 option switch |
| CPW Tank 2 | 4x push buttons, 5x 2 option switch |
| Conductivity Meter 1 | 1x push button, 1x 2 option switch |

Each Process Station will have a Human Machine Interface (HMI). The HMIs will have at least 3 system diagrams (provided by government after award). The system diagram on the HMI screen will identify the intended flow path and any active pumps. The HMI will display current tank volume levels (via analog input from the TLI's) as well as current conductivity levels.

Current software information: The CPU is programmed using Cutler Hammer GPC 5 software. The existing ladder logic consists of 546 lines of logic. All inputs and outputs are currently digital.

New PLC design parameters: PLC will be capable of receiving digital inputs (from the push buttons, and selectors). PLC will be capable of receiving analog inputs (4-20 mA) from Tank Level Indicators.